

Title

Randomized trial of a novel Staphylococcus-targeting endolysin topical therapy in mild-to-moderate atopic dermatitis: evaluation of microbial reduction and clinical efficacy

Abstract

Staphylococcal aureus overgrowth and altered skin microbiota contribute significantly to atopic dermatitis pathophysiology. Targeted antimicrobial approaches may offer therapeutic benefit. A double-blind, vehicle-controlled randomized trial evaluated a novel endolysin-based topical therapy in 53 patients with mild-to-moderate atopic dermatitis, stratified by baseline bacterial colonization status (n=26-27 per arm). Participants applied their assigned formulation twice daily for two weeks. Safety monitoring, tolerability assessment, and microbiological outcomes including bacterial recovery and microbiota composition were measured. Clinical severity scores and patient-reported symptom measures were evaluated at baseline and after two weeks. The endolysin preparation demonstrated an excellent safety and tolerability profile with no serious adverse events recorded. However, bacterial reduction did not achieve statistical significance compared to vehicle control. Similarly, clinical disease severity scores and patient-reported symptom scores showed no significant between-group differences after two weeks of treatment. Skin microbiome diversity measures remained largely unchanged. While the investigational endolysin agent demonstrated acceptable safety, short-term topical application was insufficient to produce measurable clinical or microbiological benefit in this population.